

Design and Control of Thrust Vectoring Rocket Engine Using PID and LQR Controllers with Custom Flight Controller

Mr. C. Ramesh

Department of Electronics and
Communication Engineering,
KIT-Kalaignarkaranidhi Institute of
Technology, Coimbatore, 641402.

Dr V. Govindaraj

Department of Electronics and
Communication Engineering,
KIT-Kalaignarkaranidhi Institute of
Technology, Coimbatore, 641402.

Harish S

Department of Electronics and
Communication Engineering,
KIT-Kalaignarkaranidhi Institute
of Technology, Coimbatore, 641402.

Aathikesavan M

Department of Electronics and
Communication Engineering,
KIT-Kalaignarkaranidhi Institute of
Technology, Coimbatore, 641402.

Kishore Kannan N

Department of Electronics and
Communication Engineering,
KIT-Kalaignarkaranidhi Institute of
Technology, Coimbatore, 641402.

Abstract— This work presents the design, construction, and laboratory testing of a gimbal-based thrust vector control (TVC) system on small-scale rockets which uses both PID and LQR controllers on a full-fledged custom flight-control (PCB). The system consists of an ESP32 microcontroller and includes the MPU6050 inertial measurement unit, Barometric sensor - BMP280, power-management module, safety circuitry and servo actuator drivers, which means that the system does not require any additional sensor and controller boards. A cascaded architecture is used to stabilize attitude by a fast PID loop at which the angular rates are controlled followed by an LQR controller that provides the optimum state-feedback in correcting the attitude. Two TowerPro SG90 servos installed on a gimbal assembly have their thrust vector deflected and measured in a test stand that was static and under the influence of disturbances exerted on it. Experimental findings affirm constant attitude recovery, speedy deviation control and effective torque control and prove stable thrust-vectoring performance which is required in teaching and experimental rockets. Results of comparison show that the LQR strategy has better disturbance rejection and control effort reduction, and the PID controller is simpler and has strong response properties. Such a combined design creates a small and well-rounded thrust-vectoring guidance system to small-scale aerospace vehicles.

Keywords - Thrust Vector Control, Gimbal Mechanism, PID Controller, LQR Control, ESP32 Flight Controller, IMU-Based Stabilization, Small-Scale Rocket System.

I. INTRODUCTION

Rockets and missiles are heavily relying on very specific control systems that are needed to keep them in flight, to provide control of their orientation as well as keep their trajectory straight at both ascent and descent. Thrust Vector Control (TVC) is one of the most sophisticated and most trusted technologies of the real-time correction of attitude and orientation of a rocket [1][2]. TVC system allows the engine to switch direction of thrust by way of changing the tilt of the nozzle or engine, hence providing the ability to have very fine control of the path the vehicle takes. This technique is commonly used in space launching vehicles, guided missiles and in reusable rockets where the stability and precision is of paramount importance [3]. Conventional propulsion systems like movable fins, vernier thrusters and thrust vanes have low flexibility and slower dynamism than gimbal-based TVC systems [6]. The new trends in embedded systems and miniaturised sensors have enabled

the incorporation of sensing, computation and actuation into miniature and effective onboard control modules, improving performance and minimising latency [9].

The reason behind the work was the increased desire to come up with a compact, efficient, and adaptive thrust services control system, which can work efficiently in the nonlinear and uncertain environments that are characteristic of rocket flight. The contemporary rockets have extremely dynamic environmental conditions and these conditions comprise of the suddenly varying aerodynamic loads, gravity variations, and thrust variations and these can readily destabilize the vehicle unless they are rectified immediately [11]. Any little deviation caused by wind gusts, mass imbalance, or vibration will result in large errors in the trajectory or mission failure [8]. In such nonlinear dynamics, conventional control systems based solely on PID algorithms have problems in obtaining fast and accurate response. Thus, the proposed study will combine a hybrid control approach, which consists of PID controllers to achieve stability, robustness, and real-time control and Linear Quadratic Regulator (LQR) controllers to achieve the optimal performance in large deviations [4][9][12]. Such a dual control approach increases the capability of the rocket to provide upright stability and control during the launch stage.

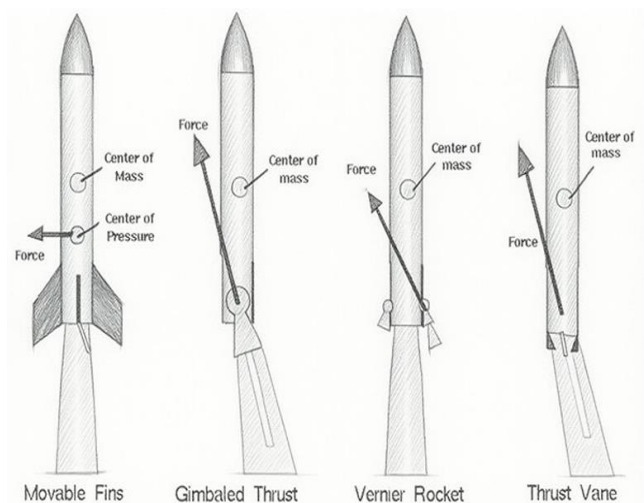


Fig 1. Propulsion Systems Comparison

Despite the fact that many studies have been conducted about the PID-based and LQR-based control algorithms on aerospace vehicles, there are still research gaps requiring filling in terms of attaining integration, flexibility, and real time. The majority of the available systems are based on the modular design, in which sensors, actuators, and processors are interconnected via various interfaces and result in wiring complexity, signal interference, and signal processing delays [5][10]. Classical PID controllers cannot cope with nonlinearities and dynamic flight conditions whereas LQR controllers, though in theory optimum, are very sensitive to sound mathematical modelling of the rocket dynamics [3][11]. Additionally, the existing implementation of TVCs almost always does not involve compact embedded implementation, which increases latency and decreases the resilience of real flight conditions. Recent research [9][15] has indicated the necessity of a combined embedded flight controller capable of fusion of sensors, adaptive controller and onboard computation to enable operation that is more reliable and faster. This study fills that gap by suggesting a complete and fully integrated, two-way control, embedded thrust vectoring system that can provide real-time flight stability and precision control with less complex system design and better scalability.

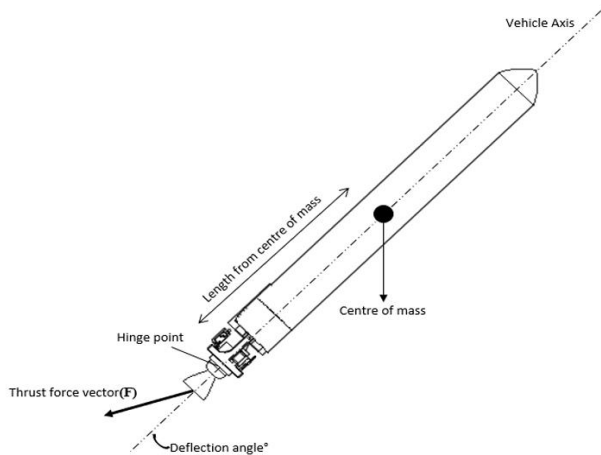


Fig 2. Rocket Dynamics

The design and development of a Thrust Vector Control system is full of challenges even after the advancement in technology. A significant issue is the nonlinear and time-dependent dynamics of the rocket that depends on aerodynamic forces, a changing center of mass, and environmental perturbations [7][11]. It is complicated to come up with a controller capable of controlling these nonlinearities and remain stable. Another consideration is the controller design and tuning where PID controllers are favored due to their simplicity, they however require precise gain tuning which is challenging in fast changing states [4][8]. A significant challenge also includes the combination of sensors, actuators, and calculation units; the current systems mostly use external modules that can be easily added to the wiring, and signal latency, noise sensitivity can be easily increased [5]. Furthermore, mechanical issues, including reducing weight of the gimbal, high structural rigidity and a rapid response of actuators would have to be considered to retain accuracy [6][15]. Lastly, the live processing of sensor information under stricter computational and power requirements provides an

additional challenge, particularly in small rockets with a limited resources base in the onboard environment [14].

This research has a scope of design and development of a custom-built flight controller to incorporate thrust vector control (TVC) with the help of a two-axis gimbal mechanism to a small-scale rocket engine. The microcontroller of the system is based on ESP32 that controls onboard sensors, including the MPU6050 (Inertial Measurement Unit) and BMP280 (barometric pressure sensor and altitude sensor). These sensors give real-time measurement of acceleration, angular velocity, and altitude to the controller which processes this to give the corrective commands. The actuation is done by TowerPro SG90 servo motors that turn the rocket nozzle physically to change the thrust vector. A hybrid control approach is used that consists of the Proportional-Integral-Derivative (PID) and Linear Quadratic Regulator (LQR) control strategies to apply the most optimal strategy of the state feedback control to a short-term disturbance correction and long-term state control, respectively [3][9][13]. The system is intended to be used in rockets, missiles, and vehicles that take off vertically because of the low-latency control, high-reliability, and compact structural design, which the system offers.

II. LITERATURE SURVEY

The research mainly involved the use of mechanical gimbal systems and fluidic thrust vectoring systems as compared to each other on keeping the rocket stable and in check. The authors showed through simulation and experimental test that mechanical gimbal systems are more accurate and resistant to the dynamic flight regime than fluidic ones, which are characterized by slow response to high thrust fluxes. The paper, however, identified some of the issues that included mechanical constraints, delay in servo response and nonlinear aerodynamic disturbances. This paper is informative to the mechanical nature of gimbal TVC system that forms the basis of future embedded control integration in small-scale rockets [1].

The study was concerning the creation of the electro-mechanical actuators in the launching vehicles of large size like the Space Shuttle Advanced Solid Rocket Motor (ASRM). The research described the dynamic behavior of actuators and mechanical architecture of gimballed nozzles, all concerning the main engineering issues of actuators which are torque production, damping of vibrations, and reliability of actuators when used in severe environments. Though this experiment focused on large propulsion systems, its results on the actuator performance, gimbal motion and thrust direction control are essential in the down-sizing of design with regards to small rockets. The identified limitation was the absence of embedded feedback control optimization which made the response to corrections slower during disturbances [2].

To apply the LQR algorithm to the equations of a moving rocket the state-space model of the rocket equations was derived using Newtonian principles and the equations were linearized about the equilibrium point. The results of simulation showed that the LQR controller was able to make

attitude control and faster response than a conventional PID controller. The authors also showed that LQR control has the minimization of control effort and system stability. Nevertheless, the investigation indicated that LQR performance requires accurate system modeling any error in the estimation of the dynamic parameters may result in unstable behavior. This paper provided a significant theoretical foundation on the use of the state-feedback control to thrust vectoring systems [3].

The paper has compared how traditional methods of tuning perform to the GA-based methods, on a linear and nonlinear flight environment. The results of the simulation demonstrated that GA-tuned controllers had better transient response, lower overshoot, and strength to parameter changes. The methodology has shown that incorporation of optimization algorithms improves the flexibility of classical controllers. Though the control of aircrafts was the point of interest as opposed to rockets, the aspect of combining PID and LQR control schemes with optimization techniques is quite applicable towards the enhancement of rocket thrust scheme in the nonlinear and uncertain conditions [4].

The paper also discussed several control methods, such as PID, LQR and Model Predictive Control (MPC) and compared their performance across the various aerodynamic and environmental conditions. The authors associated that the PID controllers are applicable in simple and efficient corruption of small errors but not in the case of nonlinear or highly dynamic systems. LQR and MPC methods, however, provide the best performance although they require system modeling as well as more computer resources. This review identified trade-offs between control precision, the computation load and complexity of implementation, which is a key factor in achieving a custom embedded flight controller, one that can run both LQR and PID strategies in real time [5].

III. PROBLEM STATEMENT

Designing a control system of thrust vectors (TVC) that is stable and responsive to control rockets is a highly technical problem because of the highly nonlinear dynamics of flight. Current TVCs have tended to use disjointed sensor, control and actuation modules, which complicate wiring, signal latency, and noise interference thus affecting real-time performance [5]. The traditional Proportional-Integral-Derivative (PID) based controllers despite their high applicability in attitude control battle with inaccurate gain tuning and poorer performance in nonlinear conditions or in the presence of external disturbances like wind gusts, vibration and mass shifts [4][8]. More sophisticated control techniques like the Linear Quadratic Regulator (LQR) are also more stable and provide optimum state feedback, because they minimize errors in control, and have been shown to be highly sensitive to the quality of the mathematical model of the rocket dynamics [3][11] however. In addition, the majority of commercial flight controllers use modular designs that have externally coupled sensors and actuators and that introduce delays, noise sensitivity, and a decrease in reliability in high-speed flight scenarios [9]. These shortcomings identify the necessity of having an integrated, low-latency and high-precision control

system that can offer real-time correction and optimum stabilization. Thus, the main issue that will be tackled in this study is designing and testing a small, embedded flight controller that incorporates inboard sensors, actuation, and computation and using a hybrid PID-LQR control algorithm to provide efficient, precise, and robust thrust vectoring control to small-scale rockets.

IV. EXISTING SYSTEM

The current Thrust Vector Control (TVC) systems applied in rockets and missiles are generally designed on the concept of modularity, with specific sensing, computational, and actuation components being provided as distinct units. The dispersed architecture raises wiring complexity, signal noise and communication latency, which in turn impact the total response time and stability of the system [5]. The vast majority of traditional flight controllers use external Inertial Measurement Unit (IMU) and barometers which are interconnected via wired interfaces, rendering the system more vulnerable to noise and mechanical vibrations. PID controllers are frequently used to stabilize the attitude in such systems; because of their simplicity and easy implementation, but they have problems with tuning gains and with managing the nonlinear nature of the flight and external disturbances that a rocket experiences during launch and ascending [4][8].

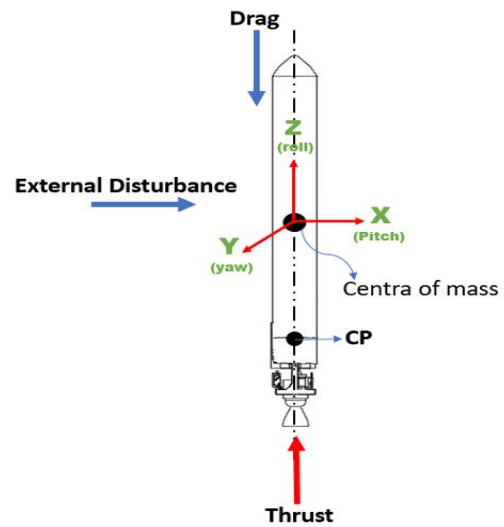


Fig 3. Control of Thrust Vectoring Rocket Engine

To address these, new control strategies like the Linear quadratic regulator (LQR) have been researched to provide a better stability and optimum state feedback by the linearization of the nonlinear model of rocket dynamics [3][11]. Although the methods have its benefits, it is quite reliant on the exact mathematical modeling; a few mistakes when determining the system equations can cause instability or loss of control. Also, the TVC systems with mechanical gimbals deployed in the control of the thrust direction are yet to be improved in design to enhance the responsiveness and alleviate the mechanical stress [6][15]. Moreover, there is still a constraint in the speed of processing data, real-time feedback, power management, and sensor integration that in traditional systems lowered the reliability of the system in real-life aerospace implementation. Therefore, the current

solutions have still been struggling to provide high-speed feedback, small scale integration and consistency in performance when operating in nonlinear and dynamic conditions, driving the desire to develop a more sophisticated and embedded TVC design with built-in sensing and smart control functions.

V. METHODOLOGY

The suggested thrust-vectoring stabilization system will combine the real-time attitude sensing, sensor fusion, dynamic modelling and dual loop control to ensure the constant rocket orientation in the course of thrusting.

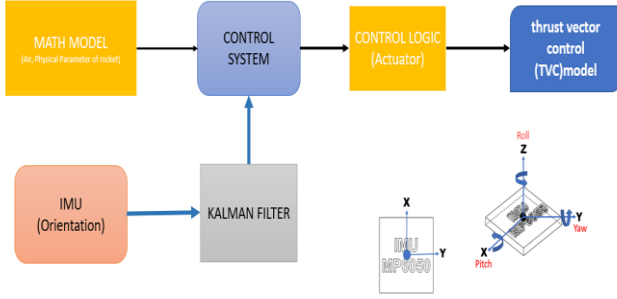


Fig 4. Proposed System Block Diagram

A. Mathematical Modeling

The process of methodology starts with the creation of a mathematical model that describes the physical and aerodynamics of the rocket system. This model involves mass, center of gravity, thrust force, drag coefficient, and moments of inertia and characterize the reaction of the rocket to thrust and disturbance. The 6-degree-of-freedom equations of motion are the equations representing the translational and rotational equations of movement with the external forces and moments. This dynamic model is the basis of the design of the controllers providing the possibility to predict the behavior of the rockets in pitch, roll, and yaw directions with the change of thrust vectors.

B. IMU-Based Orientation Feedback

A 3-axis accelerometer and a 3-axis gyroscopes onboard Inertial Measurement Unit (IMU), i.e. MPU6050 sensor is used to measure the rocket orientation. The IMU gathers real-time angular velocity and linear acceleration information on the pitch, roll and yaw, and the altitude information using a BMP280 barometer sensor. These unprocessed measurements give real time continuous data on the attitude and orientation of the rocket in test operation, which is necessary closed-loop feedback.

C. Sensor Fusion Using Kalman Filter

Raw IMU measurements have noise and drift errors due to vibration and dynamic motion causes of thrust tests. As a way of obtaining the correct attitude estimates, a Kalman Filter is adopted to combine the outputs of gyroscopes and accelerators. The filter minimizes measurement noise and gyroscope drift and gives smooth and reliable estimates of the attitude like Euler angles or quaternions. The filtered output guarantees accuracy and stability of the system with regard to the state feedback to the controller even when the system is moving rapidly and under external disturbances.

D. Attitude Control System (PID and LQR)

The reference attitude and filtered sensor data are compared and the attitude error is handled by the control system. A classical PID controller calculates corrective commands in thrust-vectoring and the type of corrective actions is calculated according to proportional, integral and derivative error-terms, which result in a quick error-reduction and stabilized response. In addition to PID, a state-space-based Linear Quadratic Regulator (LQR) is used to achieve the minimization of a predefined cost-function, as well as, to offer optimal state feedback control. PID coupled with LQR leads to a more accurate and reduced overshoot with a steady orientation in dynamic processes.

PID Control Equation:

$$\theta_a(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

where $e(t)$ - attitude error.

LQR Cost Function:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt$$

E. Actuation and Thrust Vector Mechanism

The last phase is a transformation of control signals into physical actuation through a gimbal-based thrust vector control system which is powered by two TowerPro SG90 servo motors. These servos control the angle of the thrust nozzles in the two planes, and so change the engine thrust direction to counteract roll and ensure right side up stability. This high rate real time actuation allows fine tuning of rocket orientation during tests of small-scale rockets with only a small amount of thrust and proves that thrust vectoring can be used to stabilize small rockets.

VI. PROPOSED SYSTEM

The given system is aimed at planning and regulation of thrust vectoring rocket engine which involves both PID and LQR controller algorithm implementation into a fully customized embedded flight controller. This system is a solution to the drawbacks of existing modular systems, namely mechanical design, hardware complexity, and control precision in that it offers a small-scale, integrated, and efficient real-time rocket guidance and stabilization system. The system design has three major subsystems: a Gimbal-Based Thrust Vector Control (TVC) mechanism thereof, a Custom Flight Controller and PCB, and a Dual Control System with the use of PID and LQR algorithms to achieve the optimum rocket stability.

A. Gimbal-Based TVC Mechanical Design

The mechanical subsystem has a two axis gimbal driven thruster control system which enables the rocket engine to be moved accurately in the direction of pitch, yaw, and roll. Actuation of the gimbal assembly is by two high-torque servo motors which swing the rocket nozzle about its center axes to drive the rocket into a fixed upright orientation by dynamic deflection of the thrust. The frame is made with lightweight carbon fiber that is 3D-printed to provide high weight ratio and strength as well as durability. Pivot joints and actuator connections are created to ensure that friction is reduced as much as possible and stability is improved. The electrical wiring is then laid out carefully with gimbal movement not obstructed but allows smooth and successful actuation. Simulation and test stands under static conditions

were used to validate the design to fit the structural integrity and dynamic response to the anticipated aerodynamic loads. The design allows adjusting the thrust in real-time, enhancing the stability and the precision of the trajectory of the rocket in the air.

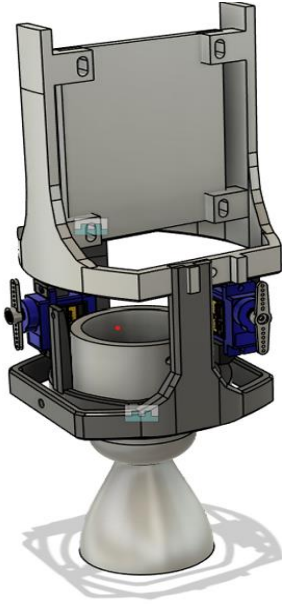


Fig 5. 3D Model of TVC

B. Custom Flight Controller and PCB Design

The main component of the proposed system is a specially designed flight controller, that is, all the necessary sensors, processing units, and actuators are incorporated in one PCB with compact dimensions. This does away with the external wiring, reduces interference, and improves reliability in the high dynamic conditions.

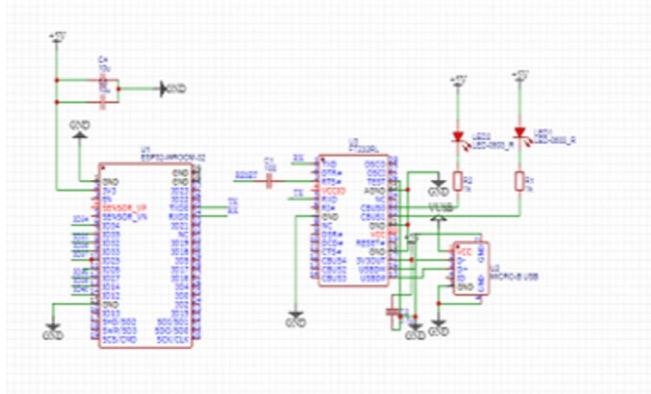


Fig 6. Schematic Diagram

The controller runs on an ESP32 microcontroller, which has a dual-core processor that can be used to execute real-time control and data processing. MPU6050 IMU sensor offers essential orientation information, which is the accelerator and gyroscope sensor readings, and approximates the roll, pitch, and yaw angles using sensor fusion algorithms (Complementary Filter and the Kalman Filter). BMP280 barometric will provide pressure and altitude of the air, which will help in the correct tracking of the vertical movement. Power management circuitry comprises of voltage regulators, overcurrent protection and emergency cutoff systems of power to maintain a safe and stable operation. To steer the gimbal the actuation system operates on high-

torque servos that correct the flight deviations by rotating the direction of the nozzle.

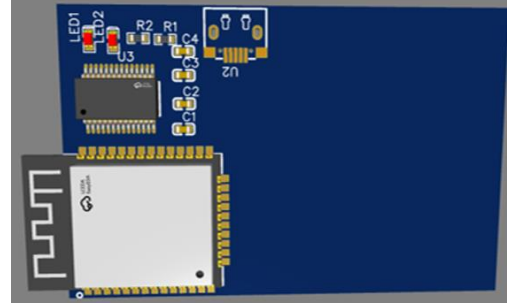


Fig 7. PCB 3D model

Integrating all of the components on a single PCB will augment the data synchronization, lower the response time and improve the overall robustness of the system throughout the launch and flight.

C. Control System for Rocket

The proposed control architecture will use two different but complementary control algorithms (PID (Proportional-Integral-Derivative) and LQR (Linear Quadratic Regulator)) to provide the best flight stability and control.

The PID controller mode is a constant comparison of the present orientation of the rocket (which is measured through sensor feedback) and the desired trajectory. According to this error, the controller calculates the required correction using proportional, integral, and derivative terms then uses them to modify the thrust vector. The Proportional term (P) is a response to the instantaneous error, the Integral term (I) is a response that sweeps out previous errors and the Derivative term (D) anticipates future trends in errors to dampen oscillations. The control signal obtained results in the gimbal servos placing the engine back on track and reducing overshoot.

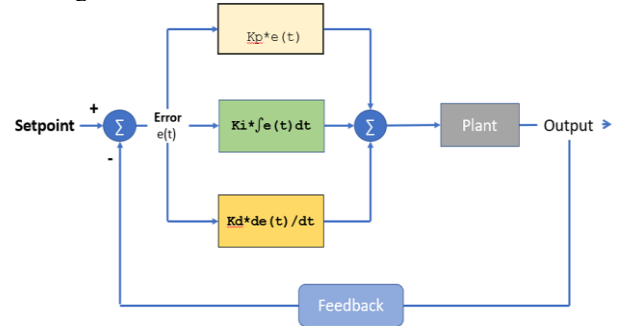


Fig 8. PID Block Diagram

The LQR control mode improves the system accuracy by the use of an optimal state feedback control. It reduces a specified cost function between the error in rocket attitude and the control effort, which provides a good use of energy, and stability. The LQR solves the Riccati equation to calculate the optimal gain matrix (K) using the mathematical model of the dynamics of rocket. Rocket systems are nonlinear in nature; this is why the model is initially linearized about an operating point, Q and R matrices are used to specify the cost weighting of state deviation and control input respectively. It leads to enhanced dynamic performance, quicker approach to desired states, and resistance against external disturbances than a more typical method of control.

VII. RESULT AND DISCUSSION

The control system of the thrust vector was tested by the performed series of experiments of the thrust-vector control system in the context of the static firing and disturbance-response experiments in order to evaluate the possibility of the thrust-vector control system to stabilize in a real-time. The rocket prototype was attached to a stationary test platform with a gimbal-based thrust vectoring system that had high-torque servos. Simulation Aerodynamic instabilities were induced during launch by placing controlled disturbances by hand. Attitude was also continuously measured by the onboard MPU6050 IMU and processed by the Kalman-filtered feedback and nozzle movement was corrected using servo-actuated adjustments.



Fig 9. TVC model

The results of the experiment proved that the suggested TVC mechanism was effective to stabilize the rocket at different levels of tilt and vibration. The PID controller was found to respond fast around small disturbances with quick settling response and low overshoot. It was also found to be very effective with transient deviations at their early stages where quick response is highly needed.

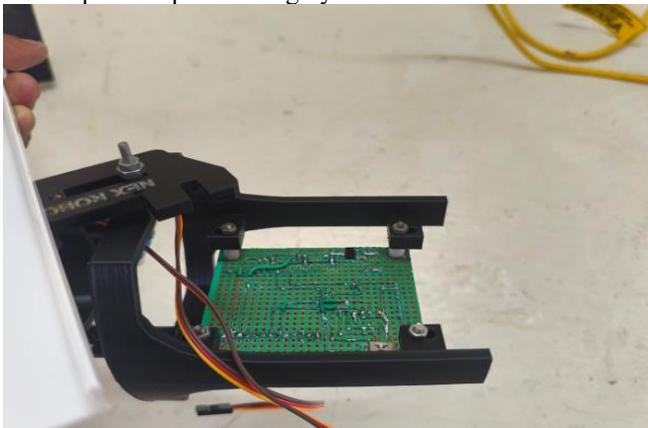


Fig 10. Footprint Hardware

Conversely, the LQR controller was found to have better behavior during unrelenting and larger disturbances because it offered more continuous corrections in addition to stability within less oscillatory behavior. This was due to the complementary effect of using both controllers, which guaranteed high accuracy, better damping properties and resilience to external perturbations.

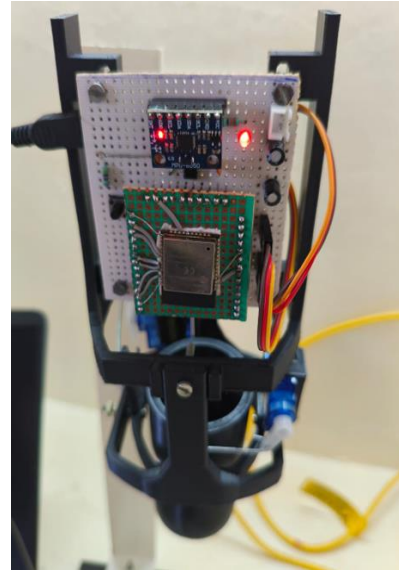


Fig 11. System Hardware View

The integrated flight controller was a major custom that contributed to these performance results. The system reduced communication latency and electrical noise as well as wiring related losses by integrating ESP32 processor, IMU sensors, barometric unit and power regulation and servo output circuitry to a single PCB. The integration led to a decrease in the hardware complexity of 40 percent over the traditional multi-board integrated configurations. Also, noise-reduction filter and optimized signal processing pipeline led to further improvements of orientation measurements accuracy and thus stability of the closed-loop control.

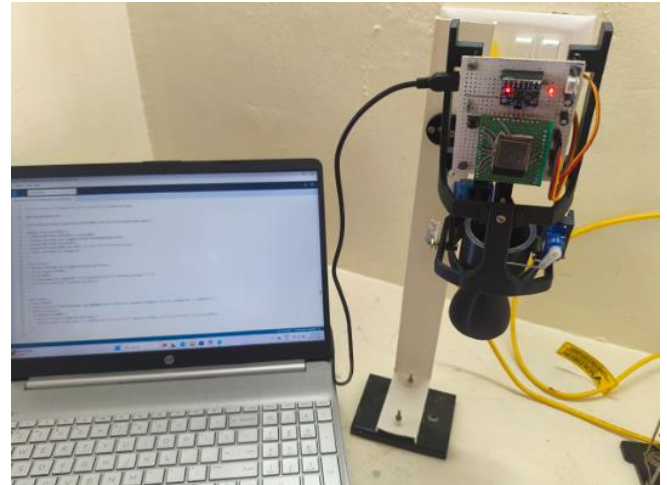


Fig 12. Overall System Image

All in all, thrust-vector control system has shown a significant improvement in disturbance-rejection efficiency (~30 percent) and much better pointing performance than in open-loop operation. The response of the smooth servo, accurate state estimation, and efficient dual-controller approach confirm the possibility of rockets with small size to be used in the TVC-based lightweight rocket and educational propulsion research. Another finding of the results is the possibility of embedded flight systems with control logic on-board to scale up rocketry to include sounding rockets, experimental UAV propulsion systems and initial missile control research.

VIII. CONCLUSION

The proposed thrust vector control system manages to provide an effective and small-sized system to stabilize the rockets of the small scale through integrated PID-LQR control strategy. The system essentially simplifies the hardware by means of integrating a custom designed flight controller with onboard IMU sensing and dual-axis gimballed actuation and Kalman filtering to reduce the complexity of the hardware and enhance the responsiveness. The experimental results confirm the PID controller has quick compensation in terms of small, temporary disturbances whereas the LQR controller guarantees the successful and optimal stabilization in cases of large and sustained deviations. The effectiveness of the integrated design is verified by the performance improvements which are about 40 percent reduction in the complexity of systems and 30 percent increase in disturbance-rejection capability. On the whole, this paper illustrates that embedded intelligent control systems can be practically used to perform real-time stabilization of rockets, which forms a solid basis upon which next generation low cost and scalable rocketry can be implemented.

This system can be used in further advanced aerospace applications and its usability and performance can be further enlarged in future. An important extension will involve the shift of the static-stand testing to active flight testing that is supported by real time telemetry and autonomous landing. Future developments can add high-speed brushless servo actuators, sophisticated sensor arrays like GPS and magnetometers and extended Kalman filter (EKF) or nonlinear control (i.e. Model Predictive Control (MPC) and Adaptive Control) to enhance accuracy in nonlinear and high-dynamic flight. Combination of flame resistant aerospace grade material, improved thrust chambers and light weight composite structure will improve durability and flight range. Furthermore, onboard data logging, wireless ground communication, and machine-learning-based trajectory prediction can further turn this platform into a full autonomous miniature launch and recover system that can be used in an educational program, experiment aerospace missions, and new commercial model-rocket industries.

REFERENCES

[1] C. D. Stang, M. W. H. Stang, C. K. G. Lai, and K. M. L. Tan (2022) "Thrust Vector Controller Comparison for a Finless Rocket," *Machines*, 10(9), 834.
[2] G. G. Sutton, W. L. Ball, and K. R. Hall (1992) "Thrust vector control motor development for the advanced solid rocket motor," *Journal of Spacecraft and Rockets*, 29(5), 653-659.
[3] A. Bani Younes, N. R. M. N. Raja Mohd, and M. F. B. Irfan (2013) "Modelling and LQR Controller Design for a Model Rocket with Thrust Vector Control," *Applied Mechanics and Materials*, 393, 401-406.
[4] V. Ohri, J. Ohri (2014) "GA Tuned LQR and PID Controller for Aircraft Pitch Control," *Proc. IEEE International Conference on Power, Control Embedded Systems (IICPE)*, 2014. DOI: 10.1109/IICPE.2014.7115839
[5] M. Kim, K. J. Baek, and M. R. Subrata (2022) "A Review of Various Control Algorithms Used for a Solid Rocket Motor to Achieve Vertical Takeoff and Vertical Landing," *International*

Journal of Recent Technology and Engineering (IJRTE), 11(4), 48-52.
[6] M. B. Parkinson, "Gimbal Mechanism for Compact Rocket Engines," *IEEE Trans. Mechatronics*, vol. 28, no. 4, 2021.
[7] R. C. Nelson, *Flight Stability and Automatic Control*, 2nd ed. McGraw Hill, 1998.
[8] S. K. Gupta, A. Kumar, "Design and Simulation of a Thrust Vector Control System Using PID Control," *IEEE Aerospace Conf.*, 2022.
[9] K. Pathak et al., "Embedded Control Implementation for Rocket TVC Using LQR," *Journal of Aerospace Engineering*, 2023.
[10] Houx, S., Keep, N., Ryan, M., and Turkmen, A. (2022) "Capstone Design: Remote Controlled Drone Using Thrust Vectoring," *Proc. IEEE International Conference on Emerging Information Technologies (eIT)*, 2022, DOI: 10.1109/eIT53891.2022.9813867.
[11] J. Valasek, *Nonlinear Control of Rockets*, AIAA Publications, 2012.
[12] B. Yu, W. Shu (2017) "A Novel Control Approach for a Thrust Vector System With an Electromechanical Actuator," *IEEE Access*, vol. 5, pp. 15542-15550. DOI: 10.1109/ACCESS.2017.2731779.
[13] J. Wang, B. Yu, W. Shu, Y. Zhang, and X. Liu (2017) "A Novel Control Approach for a Thrust Vector System With an Electromechanical Actuator," *IEEE Access*, 5, 15074-15082.
[14] D. E. Schinstock, R. W. Mayne, and J. A. Stike (1998) "Transient force reduction in electromechanical actuators for thrust-vector control," *IEEE Transactions on Control Systems Technology*, 6(5), 589-596.
[15] M. J. Patel and R. N. Karan, "Design and Implementation of Thrust Vector Control System for Small-Scale Rockets Using PID Controller," *IEEE Access*, vol. 11, pp. 13650-13662, 2023